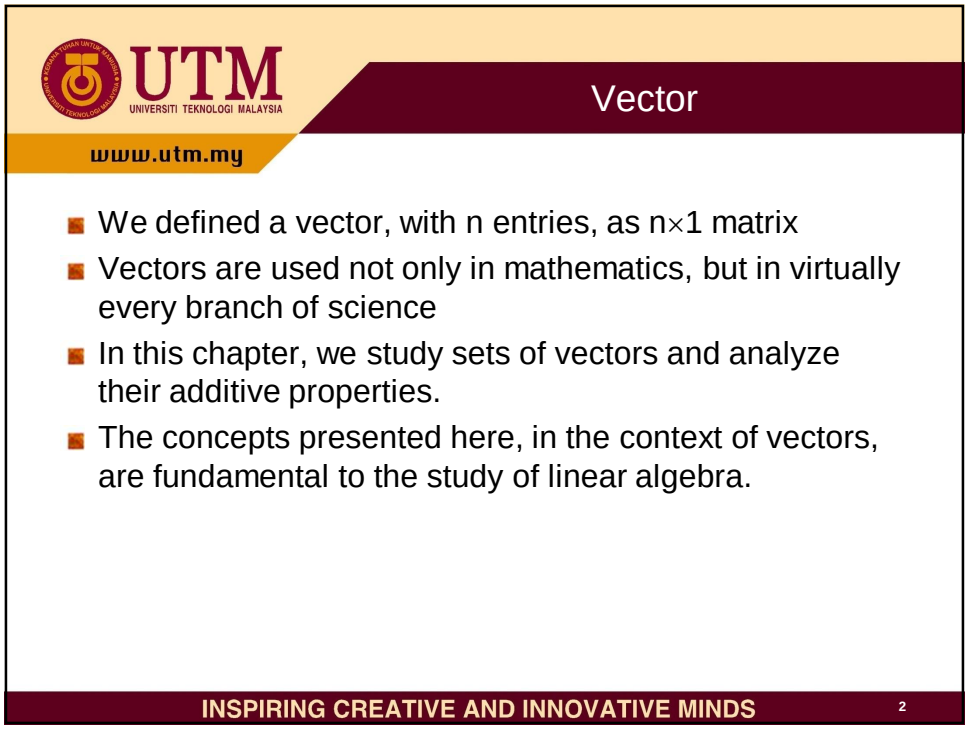

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Chapter 2
Euclidean Vector Spaces


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Vector
We defined a vector, with n entries, as n×1 matrix
Vectors are used not only in mathematics, but in virtually every branch of science
In this chapter, we study sets of vectors and analyze their additive properties.
The concepts presented here, in the context of vectors, are fundamental to the study of linear algebra.
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Vectors in R^n

- n-vector
  - If n is a positive integer, an n-vector is an n-tuple $(u_1, u_2, \ldots, u_n)$ with each coordinate u_i is a real number.
  - The set of all n-vectors is denoted R^n.

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Vectors in R^n

- R^2 consists of ordered pairs (x,y) of real numbers.
  - ***A row vector*** $v = (1,2)$
  - **A column vector** $v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
- R^3 consists of all 3-vectors or point in 3-space.

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example

- R^3

$$V=(1,2,4)$$

$$v=\begin{bmatrix}1\\2\\4\end{bmatrix}$$

- R^4

(1, 4, 3, -2) (1.5, 6, -5, π)

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Cont…

- Vectors are typically denoted with a **boldface** lower case letter. For instance we could represent the vector by **b**
- Also when we've explicitly given the initial and terminal points we will often represent the vector as, ; $\mathbf{v}=\overrightarrow{AB}$



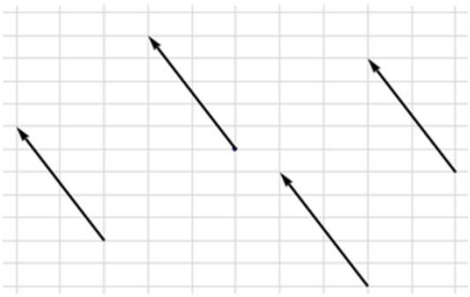
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Cont…

- Two vectors with the same magnitude but different directions **are different** vectors and likewise two vectors with the same direction but different magnitude **are different.**
- Vectors with the same direction and same magnitude are called **equivalent** and even though they may have different initial and terminal points we think of them as equal and so if **v** and **u** are two equivalent vectors we will write $\mathbf{v} = \mathbf{u}$



Equivalence vectors

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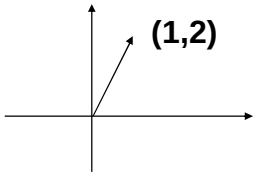
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Vector in R^n

- Geometrically, in R^2 and R^3 a vector is directed line segment from the origin to the point whose coordinates are equal to the components of the vector
- Example

$$v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

  - Directed line segment from the origin(0,0) to the point(1,2)

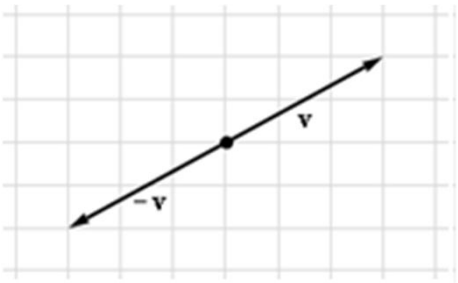


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Basic Arithmetic in Vector

- The first is the **zero vector**. The zero vector, denoted by **0**, is a vector with no length.
- If **v** is a vector then the negative of the vector, denoted by **–v**, is defined to be the vector with the same length as **v** but has the opposite direction .



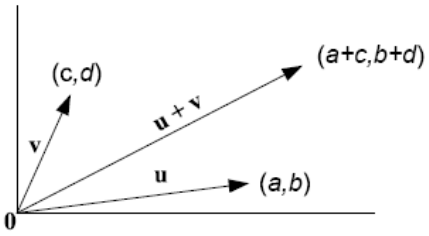
negative of a vector

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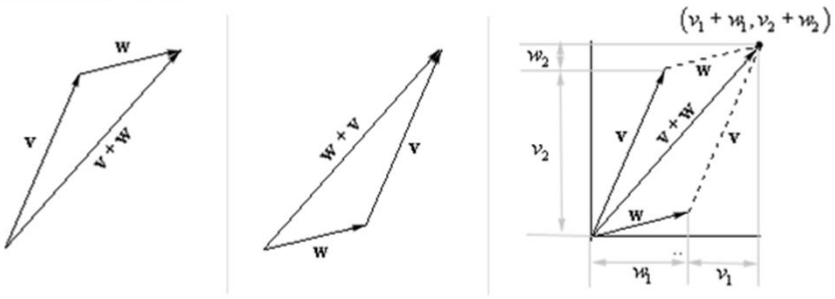
Cont…

- If (a,b) and (c,d) are the endpoint of the vectors u and v, then (a+c, b+d) will be the endpoint of u+v.



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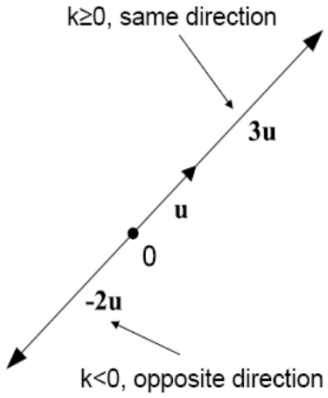
Let **v** = (v_1,v_2) and **w** =(w_1, w_2) .Then the sum of **v** + **w** can be demonstrated as the figures above.

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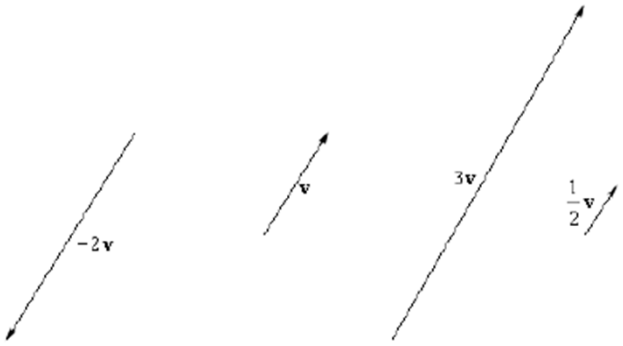
Cont…

- **Scalar multiplication**
  - The product **ku** of a real number k by a vector u is obtained by multiplying the magnitude of u by k.



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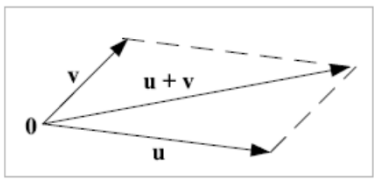
Scalar Multiples

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Vector

- **Addition**
  - The resultant **u+v** of 2 vectors u and v is obtained by the parallelogram law.
  - u+v is the diagonal of the parallelogram formed by u and v.



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Vectors in R^n

- Two n-vectors are added by adding their respective components.
- Let u and v be vectors in R^n:

$$u = (u_1, u_2, \ldots, u_n) \quad v = (v_1, v_2, \ldots, v_n)$$

$$u + v = (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$$

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Addition of Column vectors

$$u + v = \begin{bmatrix} u_1 \\ u_2 \\ . \\ . \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ . \\ . \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ . \\ . \\ u_n + v_n \end{bmatrix}$$

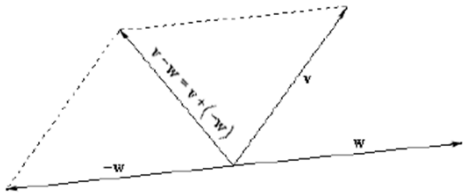
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Cont…

- Suppose that we have two vectors v and w then the difference of **w** from **v**, denoted by **v-w** is defined to be,

$$\mathbf{v}-\mathbf{w}=\mathbf{v}+(-\mathbf{w})$$



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Vectors in R^n

- An n-vector is multiplied by a scalar by multiplying each component by the scalar.
- Let u be a vector in R^n and k be a real number.

$$ku=(ku_1,ku_2,\ldots,ku_n)$$

$$ku=k\begin{bmatrix}u_1\\u_2\\.\\u_n\end{bmatrix}=\begin{bmatrix}ku_1\\ku_2\\.\\ku_n\end{bmatrix}$$

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example

- let

$$u = (1,4,5,-3) \qquad v = (8,1,-2,-1)$$

$$u + v = (1+8,4+1,5-2,-3-1) = (9,5,3,-4)$$

$$2u = (2.1,2.4,2.5,2.(-3)) = (2,8,10,-6)$$

$$2u - 3v = (2,8,10,-6) + (-24,-3,6,3) = (-22,5,16,-3)$$

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Theorem

- For any vectors u, v, w $\in R^n$ and any scalars $k_1, k_2 \in R$.
  - $(u+v)+w = u+(v+w)$
  - $u+0 = u$
  - $u+(-u) = 0$
  - $u+v = v+u$
  - $k_1(u+v) = k_1u+k_1v$
  - $(k_1+k_2)u = k_1u+k_2u$
  - $(k_1k_2)u = k_1(k_2u)$
  - $1u=u$

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Norm and Distance of Vector

- If **v** is a vector then the magnitude of the vector is called the norm of the vector and denoted by ||**v**||. Furthermore, if **v** is a vector in R^2 then,

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}$$

- and if **v** is in R^3 then we write it as

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

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Cont…

- Let u and v be vectors in R^n:

$$u = (u_1, u_2, ..., u_n) \quad v = (v_1, v_2, ..., v_n)$$

- The distance between the points u and v,

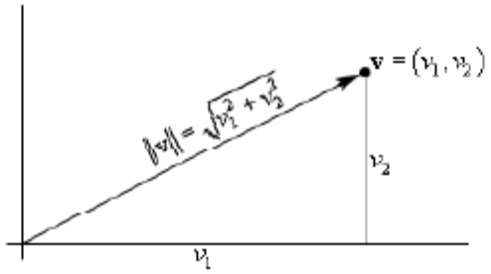
$$d(u,v) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + ... + (u_n - v_n)^2}$$

- The norm (length) of the vector u, written ||u|| is defined to be nonnegative square root of u.u:

$$\|u\| = \sqrt{u_1^2 + u_2^2 + ... + u_n^2}$$

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Norm of the vector

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Example

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- Let u=(1,-2,4,1) and v=(3,1,-5,0)

$$d(u,v) = \sqrt{(1-3)^2 + (-2-1)^2 + (4-(-5))^2 + (1-0)^2}$$
$$= \sqrt{4+9+81+1} = \sqrt{95}$$

$$\|v\| = \sqrt{3^2 + 1^1 + (-5)^2 + 0^2} = \sqrt{9+1+25+0}$$
$$= \sqrt{35}$$

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Distance of The Vector

- Let say $\mathbf{u} = (u_1, u_2, ...u_n)$ and $\mathbf{v} = (v_1, v_2, ...v_n)$. The distance between **u** and **v**, in R^n is known as Euclidean Distance and can be computed using the following formula:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + ... + (u_n - v_n)^2}$$

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example

- Suppose **u** = (2, -3, 6, 1) and **v** = (1,9, -2, 4). Find the Euclidean distance between **u** and **v** in R^4
- Solution

$$\begin{aligned} d(u, v) &= \|u - v\| \\ &= \sqrt{(2-1)^2 + ((-3)-9)^2 + (6-(-2))^2 + (1-4)^2} \\ &= \sqrt{1 + 144 + 64 + 9} \\ &= \sqrt{218} \end{aligned}$$

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Dot Product and Cross Product

- Dot products and cross product are special products of vectors. The dot product is sometimes called the **scalar product** or the **Euclidean inner product.**

- If **u** and **v** are two vectors in 2-space R^2, or 3 space R^3, and θ is the angle between them, then the dot product, denoted by u•v is defined as,

$$\mathbf{u} \bullet \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$$

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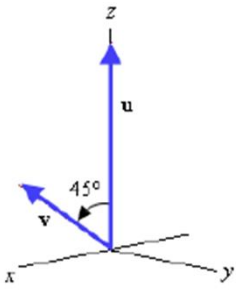
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Example

- Find the dot product for the following pairs of vectors;
  (a) **u** =(0, 0, 3) and **v** = (2, 0, 2) and θ = 45°

(a)



$$\|\mathbf{u}\| = \sqrt{0+0+9} = 3 \qquad \|\mathbf{v}\| = \sqrt{4+0+4} = \sqrt{8} = 2\sqrt{2}$$

$$\mathbf{u}\bullet\mathbf{v} = (3)\left(2\sqrt{2}\right)\cos(45) = 6\sqrt{2}\left(\frac{\sqrt{2}}{2}\right) = 6$$

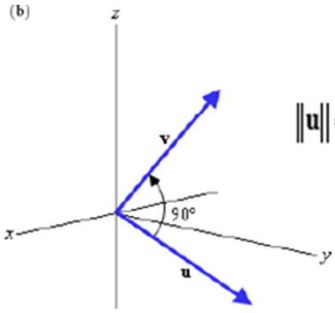
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Cont…

- (b) **u** =(0, 2, -1) and **v** = (-1, 1, 2) and θ = 90°



$$\|\mathbf{u}\| = \sqrt{0+4+1} = \sqrt{5} \qquad \|\mathbf{v}\| = \sqrt{1+1+4} = \sqrt{6}$$

$$\mathbf{u} \cdot \mathbf{v} = \left(\sqrt{5}\right)\left(\sqrt{6}\right)\cos(90) = \sqrt{30}\,(0) = 0$$

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Theorem

Suppose that $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ are two vectors in 3-space then,

$$\mathbf{u} \bullet \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$$

Similarly, if $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ are two vectors in 2-space then

$$\mathbf{u} \bullet \mathbf{v} = u_1 v_1 + u_2 v_2$$

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exercise

Given **u** = (5, -2) , **v** = (0, 7) and **w** = (4, 10) compute the following:

(a) **u●u** and $\|\mathbf{u}\|^2$

(b) **u●w**

(c) (-2**u**)**●v** and **u●(-**2**v)**

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Cross-Product

Cross product of two vectors is different with the **dot products** because the first one produces **new vectors** while the second one gives **a scalar**.

If **u** and **v** are two vectors in 3-space then the cross product, denoted by **u×v** is defined in one of three ways.

(a) $\mathbf{u}\mathbf{x}\mathbf{v} = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)$ Vector Notation

.

(b) $\mathbf{u}\mathbf{x}\mathbf{v} = \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}, -\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}, \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \right)$ using 2 x 2 determinants

.(c) $\mathbf{u}\mathbf{x}\mathbf{v} = \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$ using 3 x3 determinants

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example

- Compute **u x v** for **u** = (4, -9, 1) and **v** = (3, -2, 7).
- *Solution*: This problem can be solved using either one of the three definitions above. In this case, we will use the third one.

$$uxv = \begin{vmatrix} i & j & k \\ 4 & -9 & 1 \\ 3 & -2 & 7 \end{vmatrix}$$

$$= -63i + 3j - 8k - 28j + 2i + 27k$$
$$= -61i - 25j + 19k$$

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Exercise

- Let

$u=(3 \quad 1 \quad 2) \quad v=(-2 \quad 1 \quad 4)$

- Compute,

(a) u+v (b) **-5u**
(c) ||u|| (d) ||v||
(e) 3u–2v (f) d(u,v)

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Linear Combination

Given a set of vectors in , which have a same length of ***m***, a linear combination of these vectors is defined as the vector

- $\mathbf{v} = c_1 v_1 + c_2 v_2 + ... + c_n v_n$
- $= \sum_{i=1}^{n} c_i v_i$, where
- $c_1, c_2, ..., c_n$ are scalars.

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Example

Consider a set of vectors $v_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix}$

Can we express the following vectors $\mathbf{u} = \begin{pmatrix} 9 \\ 2 \\ 7 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} 4 \\ -1 \\ 8 \end{pmatrix}$

as a linear combinations of v_1 and v_2 ?

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solution

For the first vector **u** to be a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$ we must have

$$\begin{pmatrix}9\\2\\7\end{pmatrix} = c_1\begin{pmatrix}1\\2\\-1\end{pmatrix} + c_2\begin{pmatrix}6\\4\\2\end{pmatrix} \Rightarrow \begin{bmatrix}1 & 6\\2 & 4\\-1 & 2\end{bmatrix}\begin{pmatrix}c_1\\c_2\end{pmatrix} = \begin{pmatrix}9\\2\\7\end{pmatrix}$$

which is a linear system. The augmented matrix is

$$\tilde{A} = \begin{bmatrix}1 & 6 & 9\\2 & 4 & 2\\-1 & 2 & 7\end{bmatrix} \Rightarrow \begin{bmatrix}1 & 6 & 9\\0 & -8 & -16\\0 & 8 & 16\end{bmatrix} \Rightarrow \begin{bmatrix}1 & 6 & 9\\0 & 1 & 2\\0 & 0 & 0\end{bmatrix} \Rightarrow \begin{bmatrix}1 & 0 & -3\\0 & 1 & 2\\0 & 0 & 0\end{bmatrix}$$

Thus, $\begin{pmatrix}9\\2\\7\end{pmatrix} = c_1\begin{pmatrix}1\\2\\-1\end{pmatrix} + c_2\begin{pmatrix}6\\4\\2\end{pmatrix} = -3\begin{pmatrix}1\\2\\-1\end{pmatrix} + 2\begin{pmatrix}6\\4\\2\end{pmatrix}$

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Cont…

- For the second vector **w** to be a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$, we must have

$$\begin{pmatrix}4\\-1\\8\end{pmatrix} = c_1\begin{pmatrix}1\\2\\-1\end{pmatrix} + c_2\begin{pmatrix}6\\4\\2\end{pmatrix} \Rightarrow \begin{bmatrix}1 & 6\\2 & 4\\-1 & 2\end{bmatrix}\begin{pmatrix}c_1\\c_2\end{pmatrix} = \begin{pmatrix}4\\-1\\8\end{pmatrix}$$

- which is a linear system. The augmented matrix is

$$\tilde{A} = \begin{bmatrix}1 & 6 & 4\\2 & 4 & -1\\-1 & 2 & 8\end{bmatrix} \Rightarrow \begin{bmatrix}1 & 6 & 4\\0 & -8 & -9\\0 & 8 & 12\end{bmatrix} \Rightarrow \begin{bmatrix}1 & 6 & 4\\0 & -8 & -9\\0 & 0 & 3\end{bmatrix}$$

- Thus **w** cannot be expressed as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

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example

Show that the matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ is a linear combination of the matrices

$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $M_2 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ and $M_3 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Solution:

Firstly we must find the values of c_1,c_2 and c_3 such that

$$c_1\mathbf{M}_1 + c_2\mathbf{M}_2 + c_3\mathbf{M}_3 = \mathbf{A}$$

That is

$$c_1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + c_2\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} + c_3\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

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Cont…

Perform the scalar multiplication and addition, we get

$$\begin{bmatrix} c_1 + c_3 & c_2 + c_3 \\ c_2 + c_3 & c_1 + c_2 + c_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{aligned} c_1 + \quad + c_3 &= 1 \\ + c_2 + c_3 &= 1 \\ c_1 + c_2 + c_3 &= 0 \end{aligned}$$

Solve these equations, we obtain $c_1 = c_2 = -1$ and $c_3 = 2$. Thus, matrix A is a linear combination of matrices $\mathbf{M_1}$, $\mathbf{M_2}$ and $\mathbf{M_3}$.

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Exercise

- Determine whether the vector

$$v = \begin{bmatrix} -1 \\ 1 \\ 10 \end{bmatrix}$$

is a linear combination of the vectors

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad v_2 = \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix} \quad \textbf{and} \quad v_3 = \begin{bmatrix} -6 \\ 7 \\ 5 \end{bmatrix}$$

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Linear Dependence and Linear Independence

Consider a set of n vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, which have a same length of m. They are said to be linearly dependent if there exist n scalars, $c_1, c_2, \ldots, c_n$, not all zeros such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = 0$$

It is clear that $c_1 = c_2 = \ldots = c_n = 0$ satisfy the above equation trivially. Thus, for linear dependence, we require at least one of them to be non-zero. If say $c_1 \neq 0$, then we can write $\mathbf{v}_1 = \left(-c_2/c_1\right)\mathbf{v}_2 + \cdots + \left(-c_n/c_1\right)\mathbf{v}_n$

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Cont…

Consider a set of n vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, which have a same length of m. They are said to be linearly independent if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = 0$$

has only a trivial solution $c_1 = c_2 = \ldots = c_n = 0$.

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Properties of linearly independent sets

i) The empty set is linearly independent

ii) A set with a single nonzero vector is linearly independent

iii) A set is linearly independent $\Leftrightarrow$ the only representations of 0 as a linear combination of its vectors are trivial.

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Let

$$\mathbf{v}_1 = \begin{pmatrix} v_{11} \\ v_{21} \\ \vdots \\ v_{m1} \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} v_{12} \\ v_{22} \\ \vdots \\ v_{m2} \end{pmatrix}, \quad \cdots, \quad \mathbf{v}_n = \begin{pmatrix} v_{1n} \\ v_{2n} \\ \vdots \\ v_{mn} \end{pmatrix}$$

Then

$$c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n = \begin{pmatrix} v_{11} \\ v_{21} \\ \vdots \\ v_{m1} \end{pmatrix} c_1 + \begin{pmatrix} v_{12} \\ v_{22} \\ \vdots \\ v_{m2} \end{pmatrix} c_2 + \cdots + \begin{pmatrix} v_{1n} \\ v_{2n} \\ \vdots \\ v_{mn} \end{pmatrix} c_n = \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{m1} & v_{m2} & \cdots & v_{mn} \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = 0$$

which is an $m \times n$ homogeneous system and it has a trivial solution. If this trivial solution is the only solution, then the given vectors are linearly independent. If there are non-trivial solutions, then the vectors are linearly dependent.

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Example 1

Example: Determine whether the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 5 \\ 6 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

are linearly independent or not.

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} c_1 + \begin{pmatrix} 5 \\ 6 \\ -1 \end{pmatrix} c_2 + \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} c_3 = \begin{bmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The augmented matrix is

$$\tilde{\mathbf{A}} = \left[\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ -2 & 6 & 2 & 0 \\ 3 & -1 & 1 & 0 \end{array}\right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ 0 & 16 & 8 & 0 \\ 0 & -16 & -8 & 0 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ 0 & 1 & 0.5 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0.5 & 0 \\ 0 & 1 & 0.5 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

c_3 is a free variable

$c_1 = c_2 = -0.5c_3$

There are non-trivial solutions. Hence, they are linearly dependent.

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Example 2

- Vectors $\boldsymbol{i}=(1,0,0)$, $\boldsymbol{j}=(0,1,0)$, $\boldsymbol{k}=(0,0,1)$ in R^3

  $c_1\boldsymbol{i} + c_2\boldsymbol{j} + c_3\boldsymbol{k} = 0$

  $c_1\,(1,0,0) + c_2\,(0,1,0) + c_3\,(0,0,1) = 0$

  This implies that $c_1 = 0$, $c_2 = 0$ and $c_3 = 0$

  Thus set S= {$\boldsymbol{i}$, $\boldsymbol{j}$, $\boldsymbol{k}$} is linearly independent

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Example 3

Given $\mathbf{v_1}$ = (1, -2, 3, -4), $\mathbf{v_2}$ = (-1, 3, 4, 2), and $\mathbf{v_3}$= (1, 1, -2, -2), Determine whether the above vectors are linearly independent or not.

Solution:

$\mathbf{c_1}(1, -2, 3, -4) + c_2(-1, 3, 4, 2) + c_3(1, 1, -2, -2) = 0$

The system equation is;

$$
\begin{aligned}
c_1 - c_2 + c_3 &= 0 \\
-2c_1 + 3c_2 + c_3 &= 0 \\
3c_1 + 4c_2 - 2c_3 &= 0 \\
-4c_1 + 2c_2 - 2c_3 &= 0
\end{aligned}
$$

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Cont…

Change into matrix representation, we obtain

$$\begin{bmatrix} 1 & -1 & 1 \\ -2 & 3 & 1 \\ 3 & 4 & -2 \\ -4 & 2 & -2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving these linear equations, we found that $c_1 = 0$, $c_2 = 0$ and $c_3 = 0$. This indicates that the system has only the trivial solution and we can say that these three vectors are linearly independent.

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Exercise 1

- Determine whether the vectors

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix} \quad \text{and} \quad v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 3 \end{bmatrix}$$

are linearly independent or linearly dependent.

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Exercise 2

- Determine whether the vectors

$$\boldsymbol{v}_1=(1,0,0) \quad , \boldsymbol{v}_2=(0,1,0) \quad , \boldsymbol{v}_3=(0,0,1)$$

form a linearly dependent set or linearly independent set.